



SCHOOL OF
MANAGEMENT AND
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PAN-ATLANTIC UNIVERSITY

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DEPARTMENT OF ECONOMICS
SCHOOL OF MANAGEMENT AND SOCIAL SCIENCES
PAN-ATLANTIC UNIVERSITY, IBEJU-LEKKI, LAGOS
First Semester Examinations: 2025/2026 Academic Session

PAU-ECO 312: Data Science for Social Scientist

Number of Units: 2
Time allowed: 2 hours

Instruction: Answer question ONE compulsory, and two other questions

Question 1 (25 marks)

An economist collects student food spending data using online surveys and cafeteria transaction records at Pan-Atlantic University.

- a) Identify the type of data collected in each case. (5 marks)
- b) Mention two challenges likely to arise in this dataset. (5 marks)
- c) Suggest one data cleaning technique for each challenge. (5 marks)
- d) Explain one reason why combining these two data sources can improve the quality of economic analysis. (5 marks)
- e) Identify one ethical issue that may arise from collecting this data and suggest how it can be addressed. (5 marks)

Question 2 (20 marks)

An economist at the Central Bank of Nigeria (CBN) wants to build a model to predict next month's inflation rate using past inflation, exchange rate, and money supply data.

- a) What kind of data science use case is this? (5 marks)
- b) Name and explain two evaluation metrics that are appropriate for this type of model. (5 marks)
- c) Briefly explain what each metric measures. (10 marks)

Question 3 (20 marks)

You are building a model to predict Nigeria's stock market index (NGX) using oil prices, exchange rate, and interest rates.

- a) Which ML category does this task fall under? (4 marks)
- b) Which specific model would likely perform better and explain why (4 marks)
- c) List three common problems found in raw economic data and one technique to handle each. (12 marks)

Question 4

An economist builds a model to predict GDP growth using investment, inflation, and exchange rate data.

- a) Identify the type of machine learning used. (2 marks)
- b) Name one suitable algorithm. (5 marks)
- c) Mention one evaluation metric. (3 marks)
- d) Explain with examples the concept of confusion matrix (10 marks)

Question 5

A bank uses an AI system to detect fraudulent transactions.

- a) Explain why accuracy alone is not sufficient. (5 marks)
- b) Identify the most important evaluation metric. (5 marks)
- c) Justify your answer. (5 marks)
- d) What two visualizations would you use that best illustrate the distribution of spending patterns of students of Pan-Atlantic University? (5 marks)